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EXP.NO:2 For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

CODE:

```
import pandas as pd
data = pd.read_csv("candidate.csv")
concepts = data.iloc[:, :-1].values.tolist()
target = data.iloc[:, -1].tolist()
S = concepts[0].copy()
G = [['?' for _ in range(len(S))]]
print("Initial Specific Hypothesis (S):", S)
print("Initial General Hypothesis (G):", G)
for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(S)):
            if h[x] != S[x]:
                S[x] = '?'
        G = [g for g in G if all(g[j] == '?' or g[j] == S[j] for j in range(len(S)))]
    else:
        G_new = []
        for g in G:
            for x in range(len(S)):
                if S[x] != '?' and h[x] != S[x]:
                    new_g = ['?' for _ in range(len(S))]
                    new_g[x] = S[x]
                    G_new.append(new_g)
        G = G_new
print("\nAfter Training Example", i + 1)
print("S =", S)
print("G =", G)
print("\nFinal Specific Hypothesis:")
print(S)
```

```
print("\nFinal General Hypothesis:")
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for g in G:
```

```
    print(g)
```

OUTPUT:

```
*** Initial Specific Hypothesis (S): ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
    Initial General Hypothesis (G): [['?', '?', '?', '?', '?', '?']]

    After Training Example 1
    S = ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
    G = [['?', '?', '?', '?', '?', '?']]

    After Training Example 2
    S = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
    G = [['?', '?', '?', '?', '?', '?']]

    After Training Example 3
    S = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
    G = [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', 'Same']]

    After Training Example 4
    S = ['Sunny', 'Warm', '?', 'Strong', '?', '?']
    G = [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?']]

    Final Specific Hypothesis:
    ['Sunny', 'Warm', '?', 'Strong', '?', '?']

    Final General Hypothesis:
    ['Sunny', '?', '?', '?', '?', '?']
    ['?', 'Warm', '?', '?', '?', '?']
```